

Shortest Paths without a Map, but with an Entropic Regularizer

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The lost cow problem [Baeza-Yates et al. 1988]

- k disjoint paths of unknown length, share a starting point s
- One of the paths leads to a pasture (at unknown distance $L_{\max}$ from s)
- Cow's goal: start at s , reach the pasture; minimize total length of walk
- Deterministically: $\approx (2ek + 1) \cdot L_{\max}$ [BCR88]
- Probabilistically: $\Theta(\log k) \cdot L_{\max}$ [FFKRRV91]
(in the layered disjoint paths model; see next slide; the cow acquires more information along the way)

Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- k layered graph, edge lengths $w(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(k2^k) \cap \Omega(2^k) \cdot d_w(s, t)$ [Bur96,FFKRRV91]
- Probabilistically: $O(k^{13}) \cap \Omega(k^2/\log^{1+\epsilon} k) \cdot d_w(s, t)$ [Ram93]
(for all $\epsilon > 0$)
- Here: we improve the randomized upper bound to $O(k^2)$
(this is tight: in [BCR22] we improve the lower bound)

PY example

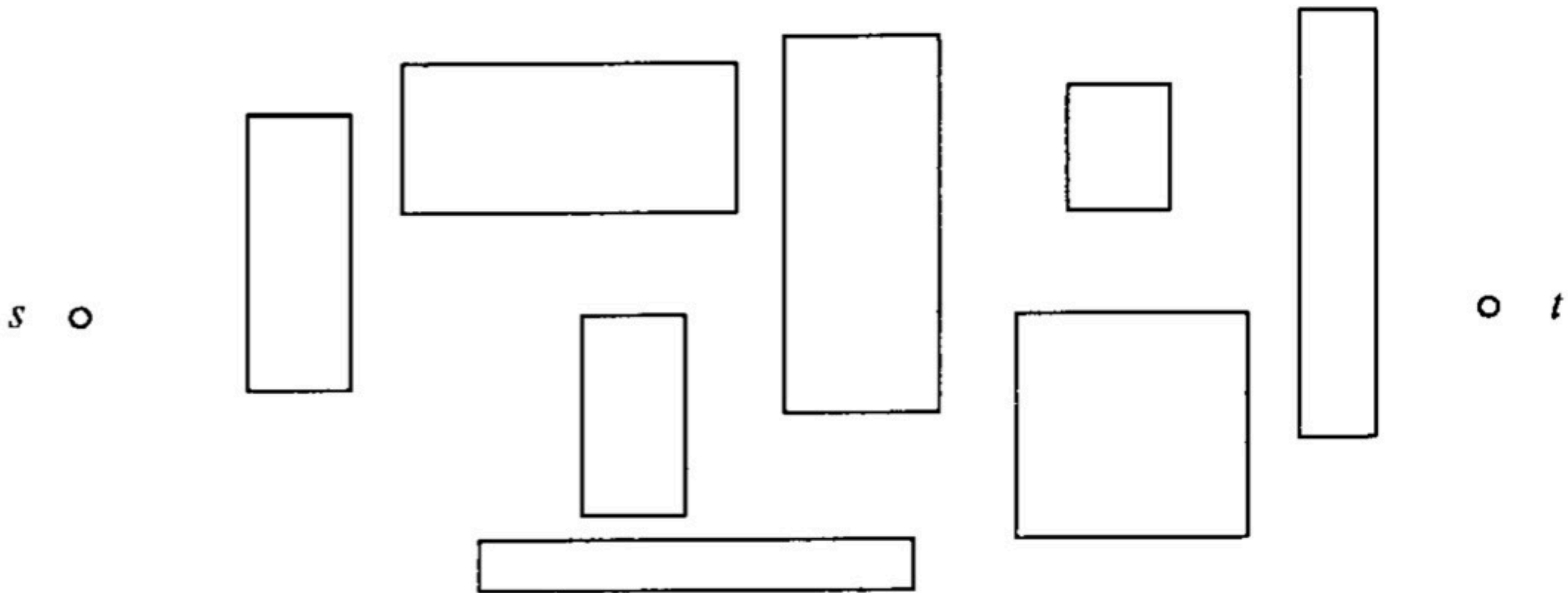


Fig. 2. Searching an obstacle scene.

Online computing (minimization problem)

- There's a two-person game between MAX and MIN, alternating moves.
- MAX pure strategies: sequences of moves, last move terminates the game. MAX has perfect recall, imperfect information (information sets group all MAX nodes reachable using same MAX strategy).
- MIN pure strategies: functions that map every alternating sequence of moves ending with a MAX non-terminating move to a MIN move. MIN has perfect information.
- At an end state: MIN cost $c_{\min}$ and base (MAX) cost c_{base} . The base cost depends only on MAX information set.
- Typical base cost:

$$c_{base}(\sigma) = A + \inf_{\alpha} c_{\min}(\sigma, \alpha)$$

Competitive analysis

- The deterministic competitive ratio: the infimum over C s.t. $\exists$ pure MIN strategy α for which $\forall$ pure MAX strategy σ ,

$$c_{\min}(\sigma, \alpha) \leq C \cdot c_{\text{base}}(\sigma)$$

- The randomized competitive ratio: the infimum over C s.t. $\exists$ mixed MIN strategy $\tilde{\alpha}$ for which $\forall$ pure MAX strategy σ ,

$$E[c_{\min}(\sigma, \tilde{\alpha})] \leq C \cdot c_{\text{base}}(\sigma)$$

- The distributional lower bound: the supremum over C s.t. $\exists$ mixed MAX strategy $\tilde{\sigma}$ for which $\forall$ pure strategy α ,

$$E[c_{\min}(\tilde{\sigma}, \alpha)] \geq C \cdot E[c_{\text{base}}(\tilde{\sigma})]$$

- Yao's minimax: randomized c.r. = distributional l.b.

Metrical task systems [Borodin et al. 1987]

- Setting: n -point metric space $\mathcal{M} = (X, d)$, initial state $\alpha_0 \in X$
- Pure MAX strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ where $\forall i$,
 $\sigma_i : X \rightarrow \mathbb{R}_+ \cup \{+\infty\}$
- Pure MIN strategies: sequences $\alpha_1, \alpha_2, \alpha_3, \dots$, where $\forall i$,
 $\alpha_i : \{\sigma_1, \sigma_2, \dots, \sigma_i, \alpha_1, \alpha_2, \dots, \alpha_{i-1}\} \rightarrow X$
- Cost:
$$c_{\min}(\sigma, \alpha) = \sum_{i \geq 1} (d(\alpha_{i-1}, \alpha_i) + \sigma_i(\alpha_i))$$
- Deterministic competitive ratio: $2n - 1$ [BLS87]
- Randomized competitive ratio: $O(\log^2 n) \cap \Omega(\log n / \log \log n)$
[BCLL19, BBM01, BLMN03]

Small set chasing [Chrobak & Larmore 1991]

- a.k.a. metrical service systems
- Setting: metric space $\mathcal{M} = (X, d)$, initial state $\alpha_0 \in X$
- Pure MAX strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ where $\forall i$,
 $\sigma_i \subset X, |\sigma_i| \leq k$
- Pure MIN strategies: sequences $\alpha_1, \alpha_2, \alpha_3, \dots$, where $\forall i, \alpha_i \in \sigma_i$
- Cost:
$$c_{\min}(\sigma, \alpha) = \sum_{i \geq 1} d(\alpha_{i-1}, \alpha_i)$$

Evolving tree game [this paper 2022]

- Setting: initial edge-weighted tree T_0 rooted at r , with one other node c_r and an edge $\{r, c_r\}$ of weight $w_{c_r} = 0$
- Pure MAX strategies: sequences $T_1, T_2, T_3, \dots$ where $\forall i, T_i$ is an edge-weighted tree rooted at r , derived from T_{i-1} by operations from:
 - Pick a leaf $\ell \neq r$ and increase the weight w_ℓ of its edge to its parent
 - Delete a leaf $\ell \notin \{r, c_r\}$ and smooth the tree at its parent (may change c_r)
 - Generate ≥ 2 new nodes and attach them to a leaf ℓ with edges of weight 0
- Pure MIN strategies: sequences $\ell_1, \ell_2, \ell_3, \dots$, where $\forall i, \ell_i$ is a leaf of T_i
- Cost: staying at ℓ while w_ℓ increases costs the increase, moving between leaves costs the current weight of the path connecting them

Reductions

k -point metrical task systems

↺

[R. Ph.D. thesis ?]

k -set chasing

↺ ↻

[FFKRRV91]

width- k layered graph traversal

↺ ↻

[FFKRRV91]

width- k layered tree traversal

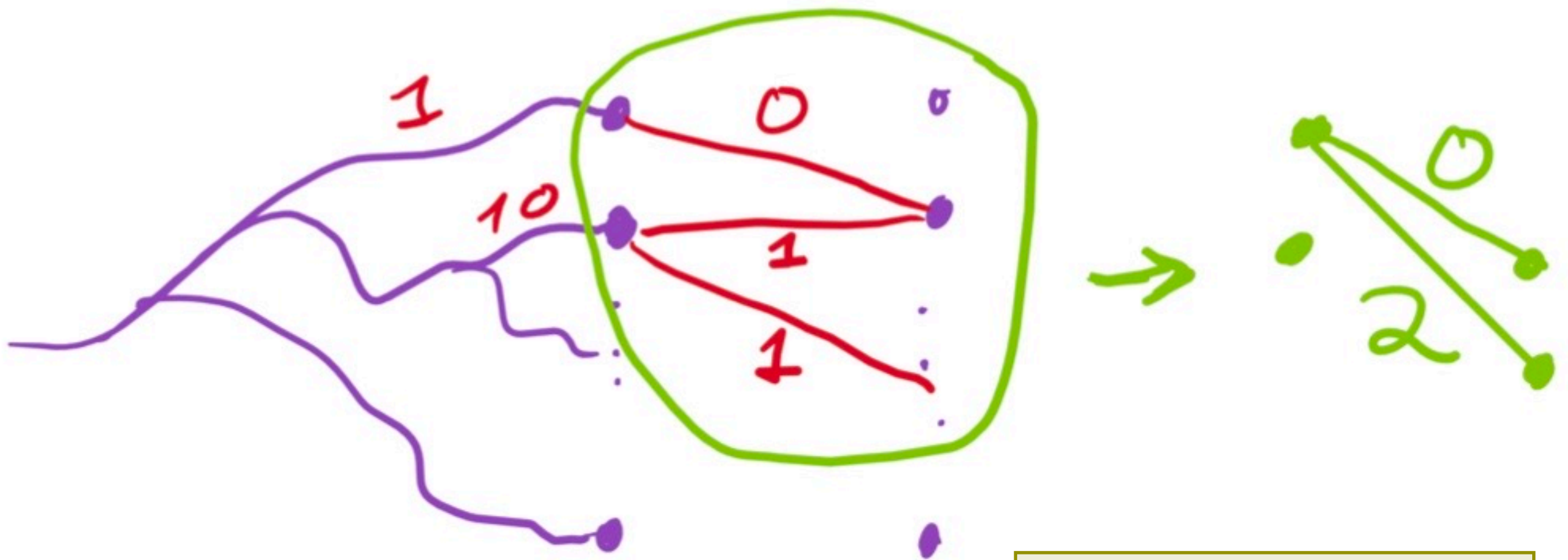
↺

[this paper]

depth- k binary evolving tree game

LGT $\propto$ LTT

Compute online the shortest paths tree



Metamorphosis

by Sébastien Bubeck
Surface Slim Pen on Surface tablet
circa 2022

Broader context: online selection

- Sequence of sets; select a point in each set; some measure of distance:
 - Convex bodies in ℓ_p^d [Sel20,AGGT20] d
(tight lower bound for $p = \infty$ [BKLLS20 ?])
 - Subsets of an n -point metric (a.k.a. MTS) [BCLL19] $O(\log^2 n)$
(faces of the n -dimensional simplex under a transportation norm)
(existentially tight lower bound [BCR22])
 - k -sets of an infinite metric (a.k.a. LGT) [this paper] $O(k^2)$
(tight lower bound in [BCR22])

Evolving tree MAX steps

- Continuous step: MAX increases w_ℓ by γ - we think of it as a continuous increase at a rate $\frac{dw_\ell}{d\tau} = 1$ for duration γ
- Deadend step: MAX deletes a leaf $\ell \notin \{r, c_r\}$
- Merge step: (after deadend step) smoothing the tree at p_ℓ if degree is 2, where $p_v \triangleq$ parent node of v
- Fork step: MAX attaches new leaves $\ell_1, \ell_2, \dots$ to a leaf u , and sets $w_{\ell_1} = w_{\ell_2} = \dots = 0$

Fractional strategies

- Fractional MIN strategy: maintain at all times a probability distribution x on the leaves of the current tree
- Extending x to internal nodes: x_v is the total probability mass at the leaves of the subtree rooted at v
- So $x_r = 1$ at all times, and $\forall u, x_u = \sum_{v \in Ch(u)} x_v$ ($Ch(u)$ = children of u)
- Cost at a leaf ℓ : $\frac{dw_\ell}{d\tau} x_\ell$
- Cost of moving from x to x' : transportation norm under w
$$= \sum_{v \neq r} w_v |x'_v - x_v|$$
- C -competitive fractional strategy $\implies C$ -competitive mixed strategy

Fractional MIN strategy

- h_v = combinatorial depth of v , we'll assume $h_v \leq k$, $\forall v$, at all times
- $\tilde{w}_v = \frac{2k-1}{2k-h_v} (w_v + \epsilon 2^{-j_v})$, where v was created in step j_v
- The term $\epsilon 2^{-j_v}$ keeps $\tilde{w}_v > 0$; we need to shift x away from 0 , too.
- Set the shift by induction: $\delta_{c_r} = 1$ and $\delta_v = \frac{\delta_{p_v}}{|Ch(p_v)|}$, $\forall v \notin \{r, c_r\}$
- This keeps $\forall v, x_v + \delta_v > 0$ (but, in fact, we'll also keep $x_v > 0$)

MIN strategy per step

- Continuous step: run, $\forall v \notin \{r, c_r\}$ in parallel, the dynamic

$$\frac{dx_v}{d\tau} = -\frac{2x_v}{\tilde{w}_v} \cdot \frac{d\tilde{w}_v}{d\tau} + \frac{x_v + \delta_v}{\tilde{w}_v} \cdot (\lambda_{p_v} - \lambda_v)$$

The (non-negative) λ s are chosen to keep x in the feasible region

Note that $\frac{d\tilde{w}_v}{d\tau} = 0$, except at $v = \ell$

- Deadend step: move to $\lim_{\tilde{w}_\ell \rightarrow \infty} x$ before deleting ℓ
(at the limit $x_\ell = 0$)
- Merge step: only the tree changes (this affects $\tilde{w}$)
- Fork step: if $q \geq 2$ children $\ell_1, \ell_2, \dots, \ell_q$ of a leaf u are created, set
 $x_{\ell_1} = x_{\ell_2} = \dots = x_{\ell_q} = x_u/q$

Analyzing MIN's strategy

- Potential function: y indicates the solution that determines c_{base}

$$P(x, y) = 4k \sum_{v \neq r} \tilde{w}_v \left(2y_v \log \frac{1 + \delta_v}{x_v + \delta_v} + x_v \right) - 2 \underbrace{\sum_{v \neq r} h_v \tilde{w}_v x_v}_{\Psi(x)} \geq 0$$

$D(x, y)$

- We need to show: $\Delta c_{\min}(x) + \Delta P(x, y) \leq C \cdot \Delta c_{base}$
- We'll show this for $C = O(k^2 \log(\max \deg))$ and $c_{base} = c_{\max}(y) + \epsilon$

Continuous step analysis

Lemma: $\frac{dc_{\min}(x)}{d\tau} = \frac{dw_\ell}{d\tau}x_\ell + \sum_{v \notin \{r, c_r\}} w_v \left| \frac{dx_v}{d\tau} \right| \leq 3 \frac{d\tilde{w}_\ell}{d\tau}x_\ell + 2 \sum_{v \neq r} (x_v + \delta_v)\lambda_v$

Lemma: $\frac{d\Psi(x)}{d\tau} \geq -k \frac{d\tilde{w}_\ell}{d\tau}x_\ell + \sum_{v \neq r} (x_v + \delta_v)\lambda_v$

Lemma: $\frac{dD(x, y)}{d\tau} \leq -\frac{d\tilde{w}_\ell}{d\tau}x_\ell + 2(2 + k \log(\max \deg)) \frac{d\tilde{w}_\ell}{d\tau}y_\ell$

Deadend step analysis

- Moving to $\lim_{\tilde{w}_\ell \rightarrow \infty} x$ costs MIN at most the drop in $P(x, y)$, by the same analysis as the continuous step (notice that $y_\ell = 0$)
- The increase in δ_v for all remaining strict descendants v of p_ℓ can only reduce $P(x, y)$, as $x_v \leq 1$ (so $\log \frac{1 + \delta_v}{x_v + \delta_v}$ monotonically non-increases in δ_v)

Merge step analysis

- Smoothing at p_ℓ doesn't increase $P(x, y)$: Write

$$P(x, y) = 2 \sum_{v \neq r} (w_v + \epsilon 2^{-j_v}) \left(\frac{2k-1}{2k-h_v} 4ky_v \log \frac{1+\delta_v}{x_v+\delta_v} + (2k-1)x_v \right)$$

h_v decreases for the nodes in the subtree rooted at the sibling of ℓ

Merging the two edges incident on p_ℓ only gets rid of the term $\epsilon 2^{-j_{p_\ell}}$

Fork step analysis

- $\ell_1, \ell_2, \dots, \ell_q$ are born in step j as children of u ,

with revised edge weights $\frac{2k-1}{2k-h_u-1} \epsilon 2^{-j} \leq \epsilon 2^{-j+1}$

$$\sum_{i=1}^q y_{\ell_i} = y_u \leq 1 \text{ and } \sum_{i=1}^q x_{\ell_i} = x_u \leq 1 \text{ and } \forall i, \delta_{\ell_i} = \delta_u/q, \text{ so:}$$

$$\begin{aligned} \Delta P(x, y) &\leq \epsilon 2^{-j+2} \cdot \left(4k \log \frac{1 + \delta_u/q}{\delta_u/q} + 2k - h_u - 1 \right) \\ &\leq \epsilon 2^{-j+2} \cdot (4k^2 \log(\max \deg) + 2k) \end{aligned}$$

- Total increase of $P(x, y)$ due to fork steps $\leq \epsilon \cdot O(k^2 \log(\max \deg))$

Motivation

- Mirror descent: strongly convex regularizer Φ , control function f ,
dynamic: $\nabla^2 \Phi(x) \frac{dx}{dt} \in f(x) - N(x)$ [NY79, BCLLM18]
 $N(x)$ is the normal cone at x of the feasible region
- The Bregman divergence $D_\Phi(y \parallel x) \triangleq \Phi(y) - \Phi(x) - \langle \nabla \Phi(x), y - x \rangle$
is a natural potential function: $\frac{dD_\Phi(y \parallel x)}{dt} \leq \langle -f, y - x \rangle$
 $-f$ is the inflicted loss, y is KAOS (has agent 86 switched sides? 🤔),
 x is the good guys _ _
- We pay also the cost of the change in x ; requires a finer analysis

Our setting

Entropic regularizer is a natural choice: $\Phi(x) = \sum_{v \neq r} w_v (x_v + \delta_v) \log(x_v + \delta_v)$

The natural control is $f(x) = -\frac{dw}{dt}$

This gives $D_\Phi(y \parallel x) = \sum_{v \neq r} w_v \left((y_v + \delta_v) \log \frac{y_v + \delta_v}{x_v + \delta_v} + x_v - y_v \right)$

So, $\frac{dD_\Phi(y \parallel x)}{dt} = y_\ell \cdot O(\log(1/\delta_\ell)) + \delta_\ell \cdot O(\log(1/\delta_\ell)) + x_\ell$

The $+x_\ell$ term can be dealt with by doubling the rate (multiply f by 2).

The $\delta_\ell \cdot O(\log(1/\delta_\ell))$ term can't be controlled, so we remove it from Φ

Movement cost

It turns out we must slow down when x_ℓ is small.

So, the revised control function is $f_\ell = -\frac{2x_\ell}{x_\ell + \delta_\ell} \cdot \frac{dw_\ell}{dt}$

Now the weighted depth potential $-\Psi(x) = -\sum_{v \neq r} h_v w_v x_v$ relates the movement cost to the loss under f (a.k.a. service cost)

But Ψ doesn't behave well under topology changes, hence $\tilde{w}$

MTS vs ETG

- MTS on a hierarchically separated n -leaf tree - c.r. $O(\log n)$:

The tree is fixed; the MAX player moves

Use conditional entropy [CL20] $\Phi(x) = \sum_{v \neq r} \frac{w_v}{\eta_v} \left(x_v + \delta_v x_{p_v} \right) \log \left(\frac{x_v}{x_{p_v}} + \delta_v \right)$

c.r. $\log \#leaves$ (doesn't depend on tree depth)

- Evolving depth- k tree game - c.r. $\Theta(k^2)$:

The tree changes; the MAX player is static;

Use uncond. entropy $\Phi(x) = \frac{1}{\eta} \sum_{v \neq r} \tilde{w}_v x_v \log(x_v + \delta_v)$ & weighted depth

c.r. $\text{tree depth} \times \log \#leaves$ (but lower bound holds even for k leaves)

Some open problems

- Randomized k -server conjecture? - we [BCR22] have a brand new $\Omega(\log^2 k)$ lower bound (specific family of $(k + 1)$ -point spaces)
- Online k -means? Best current upper bound is $O(k^2)$ [BRS21]
- Learning-augmented layered graph traversal? What's the advice?
- Efficient solution to the differential inclusion? A numerical solution requires analysis of stability.